

Zero-Dimensional Measure Extraction in Compact Spaces

Run `fa_banach_001`: consolidated stronger partial packet

18 June 2026

Source Problem and Status

The source paper is Damian Glodkowski, *A Banach space $C(K)$ reading the dimension of K* , arXiv:2207.00149, later published in *Journal of Functional Analysis*. Problem 7.3 asks:

Problem 7.3. Describe the class of compact Hausdorff spaces K such that for every non-zero Radon measure μ on K there is a zero-dimensional compact subset $L \subseteq K$ such that $\mu(L) \neq 0$.

intermediate spaces from our construction are metrizable. At that point we used the fact that for every non-zero Radon measure on metrizable compact space there is a zero-dimensional G_δ compact subset of non-zero measure (Theorem 3.16). In the light of this theorem the following problem seems to be interesting.

Problem 7.3. Describe the class of compact Hausdorff spaces K such that for every non-zero Radon measure μ on K there is a zero-dimensional compact subset $L \subseteq K$ such that $\mu(L) \neq 0$.

Assume that K is such that $C(K)$ has few operators. Then by [36, Proposition 4.8] there is a space L such that $C(K) \sim C(L)$, but $C(L)$ does not have few operators. However, by Theorem 4.19 the topology of L is very close to K , at least if we assume that K is separable and connected.

Verdict. This is a substantially strengthened partial answer, not a complete classification of all compact Hausdorff spaces. The previous packet proved finite-product and restricted-extension closure for a property that it called “strong”. The corrected result below shows that this strong-looking approximation property is actually equivalent to the original positive extraction property. It also proves countable closure, removes the unnecessary exceptional-set hypothesis from the source-paper extension step, gives ZFC nonexamples, and proves an exact characterization for arbitrary products.

1 Definitions

All compact spaces are Hausdorff, all measures are finite Radon measures, and “zero-dimensional” means covering dimension zero. For compact Hausdorff spaces this is equivalent to having a base of clopen sets.

Definition 1. A compact space K belongs to $\mathbf{E}$ if every nonzero finite positive Radon measure μ on K has a compact zero-dimensional subset $Z \subseteq K$ with $\mu(Z) > 0$.

Definition 2. A compact space K belongs to $\mathbf{E}_\delta$ if every nonzero finite positive Radon measure μ on K has a compact zero-dimensional G_δ subset $Z \subseteq K$ with $\mu(Z) > 0$.

The class E_δ is the version relevant to the countable-neighborhood extraction used in the metric part of the source paper. The class E is the literal formulation of Problem 7.3 for positive measures.

2 Extraction is Equivalent to Approximation

Theorem 1 (Extraction, approximation, and decomposition). *For a compact space K , the following are equivalent.*

1. $K \in E$.
2. For every finite positive Radon measure μ on K and every $\eta > 0$, there is a compact zero-dimensional $Z \subseteq K$ such that

$$\mu(K \setminus Z) < \eta.$$

3. Every finite positive Radon measure μ on K can be written

$$\mu = \sum_{n=1}^{\infty} \mu_n$$

as an increasing total-variation sum of positive Radon measures such that $\text{supp } \mu_n$ is zero-dimensional for every n .

Proof. Assume $K \in E$, and fix a finite positive measure μ . Let

$$s = \sup\{\mu(Z) : Z \subseteq K \text{ compact and zero-dimensional}\}.$$

Choose compact zero-dimensional Z_n with $\mu(Z_n) > s - 2^{-n}$, replacing Z_n by $Z_1 \cup \dots \cup Z_n$. Finite closed sums of zero-dimensional compacta are zero-dimensional, so Z_n remains compact zero-dimensional and the sequence is increasing. If $s < \mu(K)$, let $B = K \setminus \bigcup_n Z_n$ and define the positive Radon measure $\lambda(A) = \mu(A \cap B)$ on Borel sets. By E , there is compact zero-dimensional W with $\lambda(W) > 0$. For all sufficiently large n ,

$$\mu(Z_n \cup W) \geq \mu(Z_n) + \lambda(W) > s,$$

contradicting the definition of s . Thus $s = \mu(K)$, which proves the approximation statement.

Assume the approximation statement. Choose compact zero-dimensional $W_n \subseteq K$ with $\mu(K \setminus W_n) < 2^{-n}$, and put $Z_n = W_1 \cup \dots \cup W_n$. With $Z_0 = \emptyset$, $B_n = Z_n \setminus Z_{n-1}$, and $\mu_n(A) = \mu(A \cap B_n)$, each μ_n is a positive Radon measure and $\text{supp } \mu_n \subseteq Z_n$. Since $\mu(K \setminus \bigcup_n Z_n) = 0$, we have $\mu = \sum_n \mu_n$ in total variation.

Finally, if $\mu = \sum_n \mu_n \neq 0$, then some $\mu_n \neq 0$. Its support is compact zero-dimensional and has positive μ_n -measure, hence positive μ -measure. This gives E . $\square$

Remark 1. *The same supremum argument proves the analogous equivalence for E_δ : it is enough to replace “compact zero-dimensional” by “compact zero-dimensional G_δ ”. Finite unions of G_δ sets are G_δ .*

Proposition 1 (Signed and complex measures). *The positive-measure formulation of E is equivalent to the original Problem 7.3 formulation for real signed Radon measures, and also to the analogous formulation for complex Radon measures.*

Proof. Let ν be a nonzero signed measure. Replacing ν by $-\nu$ if necessary, $\nu^+ \neq 0$. Choose a Hahn-positive Borel set P , then by inner regularity a compact $F \subseteq P$ with $\nu^+(F) > 0$. Applying $\mathbf{E}$ inside F to $\nu^+|_F$ gives compact zero-dimensional $Z \subseteq F$ with $\nu^+(Z) > 0$, and $\nu^-(Z) = 0$. Hence $\nu(Z) > 0$. For a complex measure ν , choose θ so that $\operatorname{Re}(e^{-i\theta}\nu)$ is nonzero and apply the real case. Here we use the elementary heredity of $\mathbf{E}$ for compact subspaces, proved explicitly in the closure theorem below. $\square$

3 Metric Spaces, Closure, and the Source Extensions

Theorem 2 (Compact metrizable spaces). *Every compact metrizable space belongs to $\mathbf{E}_\delta$.*

Proof. Let K be compact metrizable, let μ be a finite positive measure, and let $\eta > 0$. Fix a compatible metric, a countable dense set (d_n) , and $D = \operatorname{diam} K$. For each n , the spheres

$$S(d_n, r) = \{x : d(x, d_n) = r\}, \quad 0 < r < D,$$

are pairwise disjoint, so only countably many have positive μ -measure. Choose distinct radii $r_{n,j}$ outside that exceptional set so that every tail $\{r_{n,j} : j \geq k\}$ is dense in $(0, D)$. Put

$$Y = K \setminus \bigcup_{n,j} S(d_n, r_{n,j}).$$

Then $\mu(K \setminus Y) = 0$. The sets $B(d_n, r_{n,j}) \cap Y$ are clopen in Y and form a base: given $y \in O$, choose $\rho > 0$ with $B(y, \rho) \subseteq O$, choose d_n within $\rho/3$ of y , and then choose a tail radius in $(d(y, d_n), \rho - d(y, d_n))$. Thus Y is zero-dimensional. By inner regularity choose compact $Z \subseteq Y$ with $\mu(K \setminus Z) < \eta$. Since K is metrizable, Z is a G_δ subset of K . Therefore $K \in \mathbf{E}_\delta$. $\square$

Theorem 3 (Countable closure). *The following closure properties hold.*

1. $\mathbf{E}$ is hereditary for compact subspaces, closed under countable closed covers, and closed under countable products.
2. $\mathbf{E}_\delta$ is hereditary for compact subspaces and closed under countable products.
3. Both $\mathbf{E}$ and $\mathbf{E}_\delta$ are closed under inverse limits indexed by a countable set, in particular by a countable ordinal.

Proof. For heredity, let $F \subseteq K$ be compact and extend a measure on F to a measure on K supported by F . Intersecting the extracting compactum with F gives the result; in the G_δ case the intersection is G_δ in F .

If $K = \bigcup_n F_n$, where the F_n are compact and belong to $\mathbf{E}$, then any nonzero positive measure on K has $\mu(F_n) > 0$ for some n , so extraction inside F_n proves $K \in \mathbf{E}$.

Let $K = \prod_{n \in \mathbb{N}} K_n$, with each $K_n \in \mathbf{E}$, and let μ be a positive probability measure on K . For the marginal $\mu_n = (p_n)_* \mu$, choose compact zero-dimensional $A_n \subseteq K_n$ with

$$\mu_n(K_n \setminus A_n) < \eta 2^{-n-1}.$$

Then $A = \prod_n A_n$ is compact zero-dimensional and

$$\mu(K \setminus A) \leq \sum_n \mu_n(K_n \setminus A_n) < \eta.$$

This gives $\mathbf{E}$. If each $K_n \in \mathbf{E}_\delta$, choose each A_n to be G_δ ; then $A = \bigcap_n p_n^{-1}(A_n)$ is G_δ in K .

Finally, a countable inverse limit of compact Hausdorff spaces is a closed subspace of their countable product. Apply product closure and heredity. $\square$

Corollary 1 (Source-paper extensions). *Let L be any extension of K by a pairwise disjoint sequence of $[0, 1]$ -valued continuous functions in the sense of the source paper. Then*

$$K \in \mathbf{E} \Rightarrow L \in \mathbf{E}, \quad K \in \mathbf{E}_\delta \Rightarrow L \in \mathbf{E}_\delta.$$

No strongness assumption and no hypothesis on $K \setminus D((f_n))$ is needed.

Proof. By the definition used in the source paper, L is a compact subspace, hence a closed subspace, of $K \times [0, 1]$. The interval belongs to $\mathbf{E}_\delta$ by the metric theorem. Product closure and heredity give the result. $\square$

Corollary 2. *Starting from a compactum in $\mathbf{E}$, respectively in $\mathbf{E}_\delta$, every countable tower formed by source-paper extensions at successor stages and inverse limits at countable limit stages remains in $\mathbf{E}$, respectively in $\mathbf{E}_\delta$.*

Remark 2. *This repairs the earlier packet's G_δ /countable-character concern for countable stages. It still does not remove all metric input from the source paper's construction: the proof of Proposition 6.2 uses a weak-star convergent subsequence of a bounded sequence of measures, and later selection arguments also use metric structure.*

4 ZFC Nonexamples

Proposition 2 (Normal-measure counterexample). *If a nondegenerate compact connected space K carries a normal probability measure, meaning a measure that vanishes on every Borel set with empty interior, then $K \notin \mathbf{E}$.*

Proof. Let $F \subseteq K$ be compact zero-dimensional. If F had nonempty interior, choose $x \in \text{int } F$ and $y \in K \setminus \{x\}$. A clopen base in F gives a nonempty set C clopen in F with

$$x \in C \subseteq (\text{int } F) \setminus \{y\}.$$

Since F is closed and $C \subseteq \text{int } F$, the set C is both closed and open in K , and it is proper. This contradicts connectedness. Thus every compact zero-dimensional F has empty interior, so a normal probability measure assigns measure zero to all such F . $\square$

Plebanek constructed in ZFC a nondegenerate compact connected space carrying a normal probability measure. Hence not every compact Hausdorff space belongs to $\mathbf{E}$.

5 Exact Characterization for Arbitrary Products

Definition 3. *A compact space K belongs to $\mathbf{S}_0$ if $\text{supp } \mu$ is zero-dimensional for every finite positive Radon measure μ on K .*

Lemma 1 (Supports). *For a positive Radon measure μ on compact K , $\text{supp } \mu$ is compact, has full μ -measure, and every nonempty relatively open subset of $\text{supp } \mu$ has positive μ -measure.*

Proof. Let U be the union of all open μ -null subsets of K . Every compact subset of U is covered by finitely many open null sets, hence is null; inner regularity gives $\mu(U) = 0$. Thus $K \setminus U = \text{supp } \mu$ is compact and has full measure. The final assertion is immediate from the definition of support. $\square$

Lemma 2. *The class $\mathbf{S}_0$ is hereditary for compact subspaces and closed under arbitrary products.*

Proof. Heredity follows by extending a measure from a compact subspace to the ambient space; the subspace support is a closed subspace of the ambient support.

Let $K = \prod_{i \in I} K_i$, with each $K_i \in \mathbf{S}_0$, and let μ be a positive measure on K . Put $\mu_i = (p_i)_* \mu$ and $S_i = \text{supp } \mu_i$. If $x \in \text{supp } \mu$, then $x_i \in S_i$ for every i , since an open μ_i -null neighborhood of x_i would pull back to an open μ -null neighborhood of x . Hence

$$\text{supp } \mu \subseteq \prod_{i \in I} S_i.$$

Each S_i is zero-dimensional. Arbitrary products of compact zero-dimensional spaces are zero-dimensional, and closed subspaces preserve zero-dimensionality. Thus $\text{supp } \mu$ is zero-dimensional. $\square$

Lemma 3 (Positive compact subsets contain full tail fibers). *Let $X = \prod_{i \in I} X_i$, where each X_i is nonempty compact and carries a strictly positive Radon probability λ_i . Let $\lambda = \bigotimes_{i \in I} \lambda_i$ be the canonical product probability. If $F \subseteq X$ is compact and $\lambda(F) > 0$, then there are a countable set $C \subseteq I$ and a point $a \in \prod_{i \in C} X_i$ such that*

$$\{a\} \times \prod_{i \in I \setminus C} X_i \subseteq F.$$

Proof. Let $U = X \setminus F$. For each n , choose compact $H_n \subseteq U$ with $\lambda(H_n) > \lambda(U) - 2^{-n}$. Cover H_n by finitely many basic open cylinders contained in U . Let V be the union of all cylinders obtained in this way. Then $V \subseteq U$, V depends on only countably many coordinates, and $\lambda(V) = \lambda(U)$.

Thus $V = p_C^{-1}(B)$ for some countable $C \subseteq I$ and open $B \subseteq X_C = \prod_{i \in C} X_i$. Put $A = X_C \setminus B$ and $P = p_C^{-1}(A)$. Then $F \subseteq P$ and $\lambda(P \setminus F) = 0$. Writing $T = I \setminus C$, let λ_C and λ_T be the corresponding product measures and let $\nu = \lambda_C|_A$. Since $\lambda(P) \geq \lambda(F) > 0$, $\nu \neq 0$. Choose $a \in \text{supp } \nu$.

The measure λ_T is strictly positive, so

$$\text{supp}(\nu \otimes \lambda_T) = \text{supp } \nu \times X_T.$$

The relatively open set $P \setminus F$ is null for $\nu \otimes \lambda_T$, hence it misses this support. Therefore $\{a\} \times X_T \subseteq F$. $\square$

Theorem 4 (Arbitrary product characterization). *Let $(K_i)_{i \in I}$ be nonempty compact spaces and put*

$$K = \prod_{i \in I} K_i.$$

Then $K \in \mathbf{E}$ if and only if both conditions hold:

1. $K_i \in \mathbf{E}$ for every $i \in I$;
2. the set $J = \{i \in I : K_i \notin \mathbf{S}_0\}$ is countable.

Proof. Assume the two conditions and let μ be a nonzero positive measure on K . For each i , put $\mu_i = (p_i)_* \mu$ and $S_i = \text{supp } \mu_i$. By the support lemma, $\text{supp } \mu$ has full measure and

$$\text{supp } \mu \subseteq P := \prod_{i \in I} S_i.$$

If $i \notin J$, then S_i is zero-dimensional. If $i \in J$, then $S_i \in \mathbf{E}$ by heredity from K_i . Since J is countable, $\prod_{i \in J} S_i \in \mathbf{E}$. The product over $I \setminus J$ is zero-dimensional, hence belongs to $\mathbf{E}$; a finite product with a zero-dimensional compactum preserves $\mathbf{E}$. Therefore $P \in \mathbf{E}$. Applying $\mathbf{E}$ to the restriction of μ to the full-measure compact set P gives the required positive-measure zero-dimensional compact subset of K .

Conversely, assume $K \in \mathbf{E}$. Each factor K_i is homeomorphic to a compact subspace of K after fixing base points in the other coordinates, so $K_i \in \mathbf{E}$. If J were uncountable, choose for each $i \in J$ a probability μ_i on K_i whose support S_i is not zero-dimensional. Regard μ_i as a strictly positive probability on S_i , and form

$$X = \prod_{i \in J} S_i, \quad \lambda = \bigotimes_{i \in J} \mu_i.$$

Let $F \subseteq X$ be compact with $\lambda(F) > 0$. The tail-fiber lemma gives a countable $C \subseteq J$ and a full tail fiber contained in F . Choose $i \in J \setminus C$. By fixing all tail coordinates except the i -th one, F contains a closed copy of S_i . Since S_i is not zero-dimensional, F is not zero-dimensional. Hence λ vanishes on every compact zero-dimensional subset of X , so $X \notin \mathbf{E}$. After fixing one base point in every coordinate $i \in I \setminus J$, the product X is homeomorphic to a compact subspace of K , contradicting heredity. Thus J is countable. $\square$

Corollary 3 (Products of compact metrizable spaces). *For nonempty compact metrizable spaces $(M_i)_{i \in I}$,*

$$\prod_{i \in I} M_i \in \mathbf{E} \iff \{i \in I : \dim M_i > 0\} \text{ is countable.}$$

In particular,

$$[0, 1]^I \in \mathbf{E} \iff I \text{ is countable.}$$

Proof. Every compact metrizable space belongs to $\mathbf{E}$. Also, a compact metrizable space M belongs to $\mathbf{S}_0$ if and only if it is zero-dimensional: if (x_n) is dense in M , then $\sum_n 2^{-n} \delta_{x_n}$ is a strictly positive probability measure with support M . Apply the product characterization. $\square$

Corollary 4. *If I is uncountable, the canonical product Lebesgue probability on $[0, 1]^I$ assigns measure zero to every compact zero-dimensional subset.*

Proof. A compact subset of positive product measure would contain a full tail fiber outside countably many coordinates. Since I is uncountable, that fiber contains a closed copy of $[0, 1]$, so the compact subset is not zero-dimensional. $\square$

Remark 3. *The cube $[0, 1]^{\omega_1}$ is a sharp obstruction to replacing countable inverse-limit closure by ω_1 -length closure: it is the inverse limit of the compact metrizable cubes $[0, 1]^\alpha$ for $\alpha < \omega_1$, but it does not belong to $\mathbf{E}$.*

Novelty and Human Review Notes

The supplied audit report states that the exact problem is repeated as Problem 3.5.3 in Glodkowski's 2024 doctoral thesis and that no later complete classification was found in searches through 18 June 2026. A quick check by the current agent also located the source paper but not a later public paper answering Problem 7.3. The packet should therefore be reviewed as a likely new strong partial answer, with the exact product theorem as the most substantive piece.

The highest-value human checks are:

1. the proof that positive extraction implies arbitrary good compact zero-dimensional approximation;
2. the tail-fiber lemma for arbitrary product Radon measures;
3. the necessity direction of the arbitrary product characterization;
4. the claim that source-paper extensions are always closed subspaces of $K \times [0, 1]$ under Definition 5.1 of the source paper.

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